

pubmed_25439526 ? erythematosus

To investigate whether serum anti-high mobility group box 1 (anti-HMGB1) antibodies are related to the development of skin lesions in systemic lupus erythematosus (SLE). This study involved 21 SLE patients with skin lesions, 18 without skin lesions, and 22 healthy controls. The presence and serum levels of anti-HMGB1-IgG and -IgM were measured by western blot and enzyme-linked immunosorbent assay (ELISA), respectively. HMGB1 expression and serum antibodies deposited in the skin were visualized by immunofluorescence staining. Using western blot analysis, we detected anti-HMGB1-IgG antibodies in 13 out of 21 SLE patients with skin lesions and 11 out of 18 SLE patients without skin lesions ($p > 0.05$). Serum levels of anti-HMGB1-IgG measured by ELISA were also comparable between the two groups of SLE patients ($p > 0.05$) but were higher in patients than in healthy controls ($p < 0.05$). Similar results were found with serum anti-HMGB1-IgM antibodies. HMGB1 accumulated under the stratum corneum in lupus cutaneous lesions without forming immune complexes with IgG or IgM, which were mainly observed along the epidermal-dermal junction. Furthermore, serum anti-HMGB1-IgM was higher in the group of patients with arthritis than in those without arthritis.

Question: Are anti-high mobility group box 1 (anti-HMGB1) antibodies related to the occurrence of cutaneous lesions in systemic lupus erythematosus?

Answer: Based on the findings of the current study, it is unlikely that anti-HMGB1 antibodies play a role in the development of SLE cutaneous lesions.